Joanna Lei

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Summary

Software Engineer focused on **AI agent** and LLM-powered applications, backend systems, and full-stack product development across Java, TypeScript, and Swift. Available to start immediately.

Education

University of Southern California Aug 2023 – Dec 2025
M.S. in Computer Science

University of California, San Diego Sep 2019 – Jun 2023
B.S. in Applied Mathematics, Minors in Business Economics and Marketing

Work Experience

Software Engineer Apr 2026 – Present
Reignite Life *Seattle, WA (Remote)*

- Designed an **agentic AI workflow** routing caregiver conversations across ALS-focused **RAG**, care-log drafting, recall, and summary flows, with clear intent boundaries and **safety-gated fallback** behavior.
- Integrated the agent backend into a **SwiftUI** client over **REST/SSE** – streaming responses with phase and replay handling, and reconciling canonical chat history to prevent duplicate messages.
- Built an **agent-driven care-log workflow** that turns user requests into structured drafts, surfaced as interactive **SwiftUI** cards for user review and confirmation before persistence.

Data Analyst May 2024 – Aug 2024
Illuminera *Shanghai, China*

- Built reusable **Python** scripts to clean, transform, and standardize user behavioral data across **5+ sources**, reducing manual preprocessing work by **70%**.
- Queried and segmented **50K+** user records with **SQL** across device, network, and user cohorts to identify retention and churn patterns, contributing to a **15%** improvement in re-engagement campaigns.
- Developed **6 interactive BI dashboards** for real-time KPI monitoring, adopted by product and marketing teams for weekly decision-making.

Project Experience

CareerForge | *Next.js, TypeScript, Vercel AI SDK, Supabase, PostgreSQL* Live demo

- Developed an AI interview-prep app with **4 tools** — resume analysis, STAR rewriting, question prediction, and multi-turn mock interviews with scorecards — streaming responses via the **Vercel AI SDK**.
- Implemented authenticated **PostgreSQL**-backed interview persistence with row-level **authorization**, enabling users to resume multi-turn interviews across sessions and review completed conversation history.

Distributed Task Queue | *Java, Spring Boot, PostgreSQL, Docker, React*

- Designed a distributed task queue in **Java/Spring Boot** with priority scheduling, exponential-backoff retries, and dead-letter routing, using row-level locking to prevent duplicate job processing.
- Built a REST API with real-time queue observability and structured JSON logging, plus an interactive **React** dashboard; deployed **5 containerized services** (API, 3 workers, Postgres) via **Docker Compose**.

Technical Skills

Languages: Java, C, TypeScript, JavaScript, Python, SQL, Swift, C++
Frameworks: Spring Boot, React, Next.js, SwiftUI, Node.js, Express.js, Tailwind CSS
Infrastructure: PostgreSQL, MySQL, MongoDB, Redis, Docker, AWS, REST APIs, SSE, WebSocket, Git, Vercel
AI & Testing: RAG, LLM API Integration, Prompt Engineering, JUnit, Postman